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RESEARCH ARTICLE

A Comparison of Approaches for Handling Concept Drifts in Data Processed With Machine Learning

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ABSTRACT In the realm of machine learning models, the pursuit of achieving favorable metrics is undeniably significant. However, these models confront phenomena that can diminish their effectiveness if left unaddressed–notably, the phenomenon of concept drift. Concept drift materializes when unforeseen alterations in the statistical properties of the target variable transpire over time. This article introduces a comprehensive analysis of classical treatment methods, examining the behavior of machine learning models across diverse datasets. Various concept drift detection algorithms are employed, facilitating a holistic assessment. The study encompasses a comparative exploration of different classification algorithms within the scikit-multiflow framework. These algorithms integrate adaptive strategies to contend with concept drifts. Through this generalized analysis, the performance of distinct classification algorithms is contrasted. The overarching aim is to facilitate the selection of optimal classification methods aligned with specific types of concept drift. Ultimately, this study provides a pivotal toolkit for the judicious selection of classification methods, enhancing model adaptability in the presence of concept drifts. In addition to shedding light on the behavior of machine learning models under concept drift, the findings empower practitioners and researchers to make informed decisions to optimize model robustness.

INDEX TERMS Classifiers, concept drift, data stream, evaluation, machine learning.

I. INTRODUCTION

With the continuous advancement of machine learning technology, it is essential to understand and address the recurring challenge in machine learning systems is essential: concept drift. Concept drift refers to the phenomenon where the statistical properties of the target variable, which the model is trying to predict, change over time in unforeseen ways [1]. This poses a critical factor that can negatively impact the performance of machine learning models trained on historical data.

The need to keep machine learning models up-to-date and free from data drift is particularly crucial in areas dealing with critical and high-risk information, such as healthcare, government data, and industries. In such contexts, the accuracy and reliability of models are vital to making accurate and impactful decisions. Adopting models that are outdated or affected by concept drift can lead to inaccurate predictions, misdiagnoses, and potentially severe consequences.

Furthermore, data generation and collection have become even more diverse and voluminous with the increasing use of Internet of Things (IoT) devices [2]. IoT devices are integrated into various spheres of modern life, including home automation, transportation, healthcare, and environmental monitoring. In this scenario, analyzing data processed by machine learning is key to deriving valuable and actionable insights from this information. However, the presence of concept drift in the data collected by IoT devices can affect the accuracy of models and consequently compromise the effectiveness of applications and services.

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In the realm of Supervised learning-based machine learning systems, which is the focus of this article, the general approach involves extracting features from the observed system. These features are used to construct data that ideally distinguishes different components of the system. The better this data can differentiate between these components, the higher the expected performance of the model during training. During the training process, this data is presented to the model, and metrics such as accuracy are analyzed to assess the system's performance. Accuracy quantifies the number of correct predictions made by the model out of the total number of samples.

However, the performance of these systems, specifically the data quality, is subject to such statistical variations defined earlier as concept drift. A practical example experienced through the concept drift phenomenon, based on its definition, was the coronavirus disease 2019 (COVID-19). Changes in global lifestyle patterns were radically altered, and models trained pre-pandemic might no longer correspond to the moment during the pandemic. The data generation during this period was a complex process undergoing substantial changes [3].

Assuming that the sample function represents the buyer's behavior for each month over a year, we can model this system as a stochastic process. From this, we can define a stochastic process as stationary, wherein a machine learning model trained at a specific time of the year would be capable of predicting customer behavior throughout the analysis period. In other words, the statistical characteristics of the random process remain unchanged over time. On the other hand, concept drift, in this sense, can be classified as a non-stationary process. The statistical characteristics of the sampled process at a given time, t_1, no longer maintain those characteristics at time $t_1 + \tau$. Therefore, these characteristics change over time. Based on this definition, it becomes necessary to develop machine learning models that are robust to concept drift, enabling the model to maintain its metrics over time.

In our paper, we will discuss different techniques that can be used to make machine learning models adaptable to the changing statistical characteristics of the underlying process. The idea is to ensure that the performance analysis metrics for models trained on data streams are not only based on traditional metrics like accuracy and F1 score but also take into account the occurrence of concept drift. Systems should be capable of detecting when concept drift occurs, where it occurs, and what actions to take to maintain model performance, making the model adaptable to this phenomenon.

The present study seeks to evaluate the impacts of concept drift on machine learning models trained in stream data, as well as strategies to mitigate these effects. Different methods of handling concept drift will be investigated, such as retraining the model with new data and using detection algorithms to identify and remove data affected by concept drift.

In this regard, it is necessary to add to this scheme of building a machine learning model three new, important methods for building more robust models. These new methods are drift detection (whether drifts occur), drift understanding (when, how, and where it occurs), and drift adaptation (how to react if drift occurs).

Concept drift covers a variety of scenarios, including gradual changes, abrupt changes, and even the presence of new concepts not previously observed in the training data. From this perspective, this article uses different methods of detecting concept drift because it is a complex task to identify the different varieties of scenarios of changes in the database accurately. Thus, it aims to vary the detection methods so that results that reflect the different scenarios of changes in the data are obtained in a real way.

Throughout this article, the need for systems with reasonable accuracies that remain in line with the dynamic behavior of the data will be emphasized. The comparative analysis will provide valuable insights for developers and researchers interested in improving their concept drift handling approaches to ensure their models' continued performance and reliability. In sum, this paper seeks to contribute to the field of concept drift handling in machine learning-processed data by offering a comprehensive overview of available approaches and identifying best practices to mitigate the negative impacts of concept drift on models. The understanding and application of these strategies are essential to ensure the effectiveness of machine learning applications in real and constantly evolving scenarios.

II. PROBLEM DESCRIPTION

This section will discuss more specific content definitions of concept drift, the different types of concept drift will be treated, once the different types of concept drift are defined, classical algorithms for detection of concept drift will be presented, and finally, the problem formulation will be discussed.

A. CONCEPT DRIFT DEFINITION

Concept drift is a phenomenon that occurs when changes in statistical properties occur with the target variable over a period of time [1]. In a more general sense, one can understand a machine learning model trained on historical data by modeling the underlying system as a stochastic process. This is akin to associating a sample function of the stochastic process with the features extracted during specific time intervals. Each sample function that composes the stochastic process can represent the characteristics of the model. According to the theory of stochastic processes, for a given time instant, t, there is a random variable that maps the process by sampling it at time t. Therefore, this random variable corresponds to points from the sample functions, meaning the features extracted at time t.

Stochastic processes have a specific classification known as stationary stochastic processes, and this definition arises from the fact that the statistical properties do not change over time. In contrast, the concept drift phenomenon can be classified as a non-stationary process [4]. Its definition lies in the fact that the statistical properties of a target variable, in this context, a given random variable X (i.e., a stochastic process sampled at time t), have different statistical characteristics for the same process sampled at a time t_2, in other words, the statistical characteristics of the random variable or target variable vary with time. This approach, in terms of a stochastic process, assumes that among the sample functions that compose it, one represents the class, and the others can be understood as feature vectors. This modeling is only at the level of problem visualization. Mathematically, concept drift is defined as a relation of conditional and marginal probabilities via Bayes' theorem.

According to paper [1], concept drift can be modeled in terms of Bayes' theorem. In other words, it involves relating the joint probability of two random variables to conditional probability and marginal probability. With that said, let the time period be [0, t], and consider a set of samples denoted as $S_{0,t} = \{d_0, \ldots, d_t\}$, where $d_i = (X_i, y_i)$ represents an observation of the samples, with X_i as the feature vector and y_i as the target variable. Hence, the joint probability of these observations X and y denotes the concept drift phenomenon if and only if:

$$p_t(X, y) \neq p_{t+1}(X, y). \tag{1}$$

In this way, the joint probability can be decomposed into the conditional and marginal probability functions. Therefore, let $p_t(X, y) = p_t(y|X) \cdot p_t(X)$.

Based on this, concept drift can be sliced into three mathematical models of occurrence. The first follows the expression $p_t(X) \neq p_{t+1}(X)$, while the conditional probability does not vary with time, classifying it as virtual concept drift. This is because if $p_t(X)$ does not affect the model's decision, based on the fact that the probability of the class being y given that X occurred is temporally invariant.

Another source of occurrence is if the marginal probability is temporally invariant, $p_t(X) = p_{t+1}(X)$, while temporal variation occurs in the joint probability $p_t(y|X) \neq p_{t+1}(y|X)$. In this case, imprecision is observed in the correct classification of a class y given the presented set of features X, leading to imprecision in the decision boundary and, consequently, a decrease in model accuracy metrics.

Finally, the combination of the other two cases of occurrence, where both conditional and marginal probabilities vary with time; in other words, concept drift focuses on the change in both $P_t(y|X)$ and $P_t(X)$.

B. TYPES OF CONCEPT DRIFT

Throughout the process discussed in this work concerning detection algorithms and the treatment methods analyzed, their precision metrics in model analyses depend on the nature of the data. This involves the intrinsic characteristics of each database, with a focus on the types of concept drift occurrences. It is from this dynamic that detection algorithms are designed and, ultimately, methods for adapting to these shifts. Thus, it is understood that some algorithms will exhibit distinct performances when encountering data whose concept drift occurrences differ from those for which the algorithm was designed to handle.

In general, there are four types, namely, abrupt, where variations in the temporal domain related to joint probabilities occur within a very short period and maintain this new configuration during subsequent times; gradual concept drift occurs over short periods, returning to old statistical properties until it persists during subsequent time instants. On the other hand, incremental concept drift, as suggested by its name, involves subtle variations related to statistical properties, taking a time window to truly perceive this variation. Finally, recurrent concept drift can be seen as abrupt but returns to past metrics and recurs temporally [5].

C. MOTIVATIONS

Several works in the scientific literature have been dedicated to robustly addressing different proposals for concept drift detection methods and robust algorithms to deal with these dynamic scenarios. However, upon conducting a meticulous bibliographic review, a gap is evident in the literature concerning the comprehensive comparison of these classification algorithms, especially those implementing concept drift treatment methods.

While some studies, such as the [6] paper, have approached the analysis of treatment methods using specific models like SVM (Support Vector Machine), these analyses often limit themselves to a few methods and a single algorithm. The study [1] explores various concepts related to concept drift, making it a significant contribution by addressing multiple aspects, such as answering questions like when, where, and how concept drift impacts a dataset. Furthermore, the paper also investigates some potential adaptation strategies for concept drift. However, it lacks results that broadly and comprehensively analyze the impact of different adaptation techniques for handling concept drift. The article [7] presents a performance analysis of several concept drift detection algorithms. The study [8] proposes algorithms for addressing concept drift specifically in the context of Global Forecasting Models. However, its scope is limited, as the proposed methods are exclusively tailored to this particular scenario. However, it does not include any examination of the different methods for handling concept drifts or how these methods relate to the results obtained. The study [9] highlights the importance of addressing concept drift by applying techniques such as model retraining, demonstrating significant accuracy improvements when treatment strategies are implemented. However, the study is limited in several aspects, including the range of treatment techniques explored, the number of datasets analyzed, and the variety of detection algorithms evaluated. Additionally, while the study provides valuable insights, its focus is primarily confined to the domain of Business Process Mining. The work [10] provides a comprehensive overview of various treatment methods, algorithms, and datasets—both real and synthetic—within the scope of its subject matter. Its primary focus is to document and summarize these aspects as presented in various studies from the literature, offering significant insights and observations. However, it should be noted that the study does not include a comparative analysis of these methods or datasets.

Unlike all the works cited, this is the first study to compare the performance of different concept drift treatment techniques in a generalized manner, encompassing a diverse range of both artificial and real-world datasets affected by various types of concept drift occurrences. It also incorporates different detection algorithms and analyzes classification algorithms in conjunction with treatment methods. Moreover, it highlights the critical importance of these treatment techniques in a comprehensive manner.

This article's proposal goes beyond, aiming for a thorough evaluation of treatment methods on diverse datasets, considering a variety of classification algorithms. Understanding the behavior of machine learning models in varied scenarios regarding datasets, detection algorithms, classification algorithms, and treatment methods motivates this work to contribute to the field of data drift by providing extensive information about these comparisons.

The distinctive aspect of this work lies not only in comparing concept drift treatment methods but also in proposing innovative approaches and ensemble strategies. These methods are applied across various concept drift scenarios, broadening the understanding of their effectiveness in different contexts and with diverse algorithms. This significantly contributes to advancing knowledge in concept drift treatment, addressing a gap identified in the existing literature. By adopting a comprehensive and comparative approach, this study aims to provide valuable insights for researchers, practitioners, and machine learning enthusiasts. It offers a deeper understanding of how models behave in the face of dynamic changes in data, thereby driving progress in the application of these techniques to real-world scenarios.

III. EVALUATION OF TREATMENT METHODS

This section initially analyzes classical methods for treating concept drift individually. Subsequently, the behavior of a machine learning model is examined using two classifiers: one without concept drift adaptation and another that performs concept drift detection and adaptation. Furthermore, a broader comparison is conducted by contrasting different classifiers incorporating concept drift detection and adaptation.

Across all the aforementioned comparisons, a comprehensive and generalized analysis is undertaken, encompassing variations in diverse datasets, concept drift detection algorithms, and classifiers. This approach ensures a conclusive evaluation. Through these examinations, insights into the efficacy of distinct treatment strategies are garnered, offering a comprehensive perspective on the interplay between concept drift and machine learning models. The results contribute to the informed selection of optimal classifiers, optimizing their performance under diverse concept drift scenarios.

A. METHODOLOGY

In this subsection, the methodology employed for the comparison of concept drift treatment methods in this study will be outlined. We will elaborate on the classification algorithms utilized, as well as the datasets employed. Furthermore, we shall present the parameters selected for the machine learning methods and explain the evaluation metrics employed. This comprehensive description serves to establish a solid foundation for the subsequent analyses and findings.

1) DATASET

Eight distinct datasets were employed to conduct a comprehensive analysis encompassing various scenarios of concept drift occurrence. This approach aims not only to scrutinize the behavior of classifiers in the presence of concept drift but, more importantly, to evaluate the classifiers that best accommodate different types of concept drift occurrences. By doing so, the objective is to present outcomes that could prove valuable across diverse situations.

- **Airlines**: Real-world data containing information from scheduled departures of commercial flights within the US. The objective is to predict if a flight will be delayed [11].
- **Covertype**: This data set contains data collected over time by the US Forest Service. Classes correspond to cover type in a forest on squares of 30×30 meters [12].
- **Electricity Market**: Data from the Australian New South Wales electricity market where prices are not fixed but change based on offer and demand. The 2 target classes represent changes in the price ($1 =$ up or $0 =$ down) [13].
- **SEA-abrupt and SEA-gradual**: A data stream with three numerical features where only two attributes are related to the target class. Created using the SEA generator. Three abrupt drifts are simulated for SEAa and three gradual drifts for SEAg [11].
- **Hyperplane-fast**: A data stream with fast incremental drifts where a d-dimensional hyperplane changes position and orientation. Created with the random hyperplane generator [11].
- **Moving Squares**: Four equidistantly separated, squared uniform distributions are moving in the horizontal direction with constant speed. The direction is inverted whenever the leading square reaches a predefined boundary. Each square represents a different class. The added value of this dataset is the predefined time horizon of 120 examples before old instances start to overlap with current ones. This is especially useful for dynamic

sliding window approaches, allowing to test whether the size is adjusted accordingly [14].

- **Weather**: Contains weather information collected between 1949 and 1999 in Bellevue, Nebraska. The objective is to predict if it will rain or not on a given date [15].

The table 1 summarizes the characteristics of the databases used in the experiments:

2) FRAMEWORK DESCRIPTION

It is well known that data streams are generated at a considerably high speed, which poses challenges in processing and, for instance, storing this data. This is due to the rapid generation speed, coupled with substantial memory consumption, making it impractical to store and apply machine learning approaches to historical data. Furthermore, many companies and governmental entities adopt a hybrid model, where new data, to be analyzed using a pre-trained model, needs to be integrated. As mentioned earlier, these scenarios bring about certain challenges and complexities.

In other words, these algorithms must be modified to cater to limitations in resources and challenges stemming from factors such as single-pass processing, memory constraints, rapid data access, and the occurrence of concept drift. This adaptation is essential due to the dynamic characteristics of data streams as they evolve over time.

In this context, this article relies on the analyses conducted within the framework of the scikit-multiflow library. scikit-multiflow represents an open-source machine learning toolkit adapted to deal with streaming data. It expands on existing Python scientific tools, catering to scenarios where data is generated continuously and requires real-time processing and analysis. Unlike traditional approaches, this framework does not store data samples and instead exposes learning techniques to new data instances only once [16].

The inherent infinite nature of data streams introduces additional complexities. Although the flow of data is limitless, resources such as memory and time remain limited, issues discussed earlier which force stream learning methods to be efficient. Furthermore, the dynamic nature of the environment means that data characteristics can evolve over time. In the scikit-multiflow library, it is possible to work with different methods for detecting concept drift and providing various types of classifiers and regressors, thus providing the essential tools for the study covered in this article.

The Table 2 provides a detailed description of the concept drift detection algorithms utilized in this study.

On the other hand, the descriptions of the classification algorithms evaluated are as follows.

a: HOEFFDING TREE (HT)

HT is a classification algorithm based on incremental decision trees capable of learning from a substantial stream of data. It assumes that the distribution of examples remains unchanged over time; therefore, the algorithm does not employ adaptation techniques for handling concept drift. It is a tree-based algorithm.

b: HOEFFDING ADAPTIVE TREE (HAT)

Contrasting with the HT algorithm, the HAT algorithm employs the concept drift detection algorithm ADWIN to monitor performance within the decision tree branches. If the evaluated classification metrics undergo a performance degradation stemming from factors such as concept drift, the algorithm implements a treatment technique by substituting the old branches with new ones, provided that these new branches exhibit better accuracy [22]. It is a tree-based algorithm.

c: EXTREMELY FAST DECISION TREE (EFDT)

EFDT is also a classification algorithm based on an incremental decision tree structure. The algorithm possesses the capability of swift learning for a given set of stationary data. Consequently, due to its reliance on a stationary dataset, the algorithm refrains from applying treatment methods in the event of concept drift occurrences. The EFDT aims to identify the optimal split during the model training process. If a particular split is deemed beneficial, the algorithm revisits the choice to pursue an even more advantageous division [23]. It is also a tree-based algorithm.

d: K-NEAREST NEIGHBORS ADAPTIVE WINDOWING (KNNADW)

KNNADW is a classification algorithm based on the classical KNN (K-Nearest Neighbors) classifier. KNNADW is a version of KNN designed for learning environments subject to concept drift occurrences. In other words, the distinction lies in its utilization of concept drift treatment methods. To achieve this, the concept drift detection algorithm ADWIN is incorporated. Once a drift occurrence is identified, the algorithm discards or removes the samples, adjusting the window size. It is a K-Nearest Neighbor-based algorithm.

e: ACCURACY WEIGHTED ENSEMBLE CLASSIFIER (AWEC)

AWEC is a classification algorithm based on ensembles, which means it's a collection of classification models where each model's classification accuracy in a continuous data stream environment determines its weighting. Depending on their individual performance, the models are weighted accordingly. This ensemble ensures robustness and efficiency in scenarios involving concept drift occurrences. For instance, the learning models used in the ensemble can incorporate concept drift handling techniques [24].

f: ADAPTIVE RANDOM FOREST CLASSIFIER (ARFC)

ARFC is a classification algorithm based on the random forest approach, and one of its important features is the utilization of concept drift detection algorithms for each decision tree that comprises the classifier. This allows for selective reinitialization in response to concept drift

TABLE 1. Dataset description.

Dataset	Type	Total of samples	Real/Artificial
Airlines	Binary classification	539,383	Real Dataset
Covertype	Multi-class classification	581,012	Real Dataset
Electricity	Binary classification	45,312	Real Dataset
Weather	Binary classification	18,159	Real Dataset
SEA-abrupt	Binary classification	1,000,000	Artificial Dataset
SEA-gradual	Binary classification	1,000,000	Artificial Dataset
Hyperplane-fast	Binary classification	1,000,000	Artificial Dataset
Moving Squares	Multi-class classification	200,000	Artificial Dataset

occurrences. Additionally, it distinguishes itself by ensuring diversity in the model construction process. This is achieved through both the random selection of subsets for node splitting in decision trees and the resampling process [11].

g: VERY FAST DECISION RULES CLASSIFIER (VFDRC)

The VFDRC is an incremental classifier that shares some similarities with the HT classifier. Unlike a decision tree-based classifier, it employs a collection of rules during the model training process, called an incremental rule-learning algorithm. The algorithm is highly interpretable, like decision trees in its model construction methodology. However, the algorithm does not implement adaptation techniques for scenarios involving concept drift occurrences [25].

h: ADDITIVE EXPERT ENSEMBLE CLASSIFIER (AEEC) OR ALSO (ADDEXP)

AEEC is an ensemble-based classification algorithm, which is a general method applied to any online learning problem. The classifier is designed to handle concept drift, making it an essential tool for working with continuous data streams subject to concept drift occurrences [26].

i: BATCH INCREMENTAL ENSEMBLE CLASSIFIER (BIC)

The BIC is an incremental ensemble classifier. The classifier incrementally constructs a set of model samples in batches for training. The algorithm uses a sample window to train a model, and in this process, new batches are added to the ensemble. In a continuous data stream environment, to ensure limited memory consumption, a maximum number of models comprise the ensemble. When this number of models is exceeded, the algorithm removes the older training models, keeping them up-to-date with the most recent models.

B. RESULTS

To analyze the performance of different classification and detection algorithms implemented in this work, the use of default parameter values was motivated by the objective of ensuring a fair and impartial comparison among classifiers. During training, the default parameters for classification algorithms were used, considering a batch size of 10,000 and a pretrain size of 1,000. Furthermore, for all datasets analyzed, the detection algorithms also utilized their default parameter values. By adhering to these default configurations, we ensure that the observed performance differences are primarily attributed to the concept drift handling strategies rather than being influenced by parameter optimization. This approach reflects our commitment to maintaining the integrity and validity of the comparative analysis.

In summary, our methodology encompasses the selection of diverse datasets, the incorporation of concept drift detection algorithms, the utilization of classification algorithms with default parameter values, and a rationale prioritizing rigorous and unbiased comparison. These methodological considerations collectively form the foundation of our investigation into the effectiveness of concept drift handling strategies in continuous data streams.

Importantly, parameter tuning to achieve better accuracy was intentionally avoided, as the focus of this analysis lies in examining the performance of classification algorithms under the presence of concept drift rather than aiming for the highest possible accuracy. This is in contrast to another analysis, detailed in the Results section, where specific concept drift handling methods are evaluated on two distinct classifiers. In this scenario, one of the classifiers employed is KNN, configured with the following parameters: The number of nearest neighbors was set to 8, the maximum size of the window for storing the last observed samples was set to 2,000 and the maximum number of samples that can be stored in a leaf node, which determines the point at which the algorithm switches to a brute-force approach, was set to 40. This dual-level analysis highlights the adaptability of the proposed methods and offers insights into their performance across diverse configurations.

The results presented in this work are divided into three stages. The first stage involves the implementation of concept drift handling techniques separately from the employed classification algorithms. In other words, for a given classifier, a concept drift detection algorithm is employed. If this algorithm detects any suspicious change, a technique is applied, which is described as follows:

- **Ignore**: If a concept drift is identified by the detection algorithms, no action will be taken.
- **Delete**: Samples identified with concept drift by the detection algorithms will be excluded.

TABLE 2. Concept drift detection algorithms.

Algorithm	Description
ADWIN	ADWIN(ADaptive WINdowing) [17] is a concept drift detection algorithm based on data stream windowing. In essence, the algorithm analyzes statistical changes between two sliding windows that continuously update within the data stream. To achieve this, the algorithm determines the width of the window by segmenting the data stream into distinct points and subsequently examining the average statistical metrics between these two windows. If the absolute difference between these average metrics surpasses a certain threshold, the algorithm identifies the occurrence of concept drift. Concept drift is characterized by a shift in statistical metrics between the two windows.
DDM	DDM (Drift Detection Method) [18] is a concept drift detection algorithm based on the PAC learning model's premise. According to this model, if the example distribution remains stationary, the learning algorithm's error rate will decrease as the number of examples increases. As a result, a notable increase in the algorithm's error rate indicates the presence of concept drift or implies its forthcoming occurrence. In other words, when the algorithm's error rate surpasses a certain threshold, the detection of concept drift or its imminent occurrence is flagged, referred to as a warning zone.
EDDM	EDDM (Early Drift Detection Method) [19] is a concept drift detection algorithm that incorporates concepts from the DDM algorithm while introducing techniques to identify a wider range of drift types. While classic algorithms like DDM yield good results in abrupt changes within the data flow, they encounter challenges in identifying slow and gradual changes. Such gradual changes cause a gradual increase in the algorithm's error rate, rendering their detection intricate. EDDM, on the other hand, employs the average distance between two classification errors. In contrast to the DDM algorithm, which relies solely on error rate metrics, the EDDM algorithm demonstrates an improved ability to detect slower gradual changes.
HDDM_A	HDDM_A (Drift Detection Method based on Hoeffding's bounds with moving average-test) [20] is a concept drift detection algorithm that incorporates enhancement techniques within its framework. The algorithm is founded on Hoeffding's inequality, utilizing the mean as an estimator. For a given stream of input data, the algorithm assesses the state of the data flow. This evaluation is based on detecting variations in the population mean through monitoring differences in moving averages. The algorithm then provides information on its current state, which can be categorized as stable, within a warning zone, or indicating the detection of concept drift.
Page-Hinkley	Unlike certain concept drift detection algorithms discussed earlier, the Page-Hinkley algorithm [21] does not provide alerts regarding the alert zone's status. This method only indicates the presence or absence of concept drift. Typically, this method identifies the presence of concept drift when the observed average exceeds a predefined lambda threshold at a specific time. In other words, concept drift is detected when the observed average surpasses the lambda threshold at a given time.

- **Retrain**: In case concept drift is identified, the system will be retrained with all the samples.
- **Batch Training**: Training is performed based on a package of samples, a subset of the total samples, aiming to adapt to the concept drift.

The second stage utilizes classification algorithms that internally implement techniques for detecting and adapting to concept drifts. It compares these classification algorithms with algorithms that do not implement this treatment approach. Finally, the third stage of results presents comparisons between various classifiers that implement concept drift handling techniques. Some of these classifiers may not necessarily have techniques explicitly stated in their formulation but are based on robust training to concept drift deviations. This analysis aims to observe the best algorithms in terms of performance by examining the algorithms and

TABLE 3. Accuracy of the models when performing different treatment methods on the SEA-abrupt dataset.

Handling methods	ADWIN	DDM	EDDM	HDDM_A	PageHinkley
Ignore	0.864168	0.864168	0.864168	0.864168	0.864168
Delete	0.864168	0.864168	0.864168	0.864168	0.864168
Retrain	0.864165	0.864169	0.864163	0.864166	0.864205
Batch training	0.869518	0.869518	0.869518	0.869518	0.869518

TABLE 4. Accuracy of the models when performing different treatment methods on the SEA-gradual dataset.

Handling methods	ADWIN	DDM	EDDM	HDDM_A	PageHinkley
Ignore	0.807275	0.807275	0.807275	0.807275	0.807275
Delete	0.807275	0.807275	0.807275	0.807275	0.807275
Retrain	0.806543	0.807265	0.806706	0.806483	0.80793
Batch training	0.8348289	0.8348289	0.8348289	0.8348289	0.8348289

TABLE 5. Accuracy of the models when performing different treatment methods on the Electricity Market dataset.

Handling methods	ADWIN	DDM	EDDM	HDDM_A	PageHinkley
Ignore	0.7828169	0.7828169	0.7828169	0.7828169	0.7828169
Delete	0.7828169	0.7828169	0.7828169	0.7828169	0.7828169
Retrain	0.7834569	0.7828169	0.776902	0.77679	0.780036
Batch training	0.79668	0.79668	0.79668	0.79668	0.79668

TABLE 6. Accuracy of the models when performing different treatment methods on the SEA-abrupt dataset (KNN).

Handling methods	ADWIN	DDM	EDDM	HDDM_A	PageHinkley
Ignore	0.686247	0.686247	0.686247	0.686247	0.686247
Delete	0.686247	0.686247	0.686247	0.686247	0.686247
Retrain	0.686245	0.686243	0.686249	0.686249	0.686239
Batch training	0.7694889	0.7694889	0.7694889	0.7694889	0.7694889

the data distribution. It may provide insights to identify the best algorithms depending on data distribution, data sampling/collecting periods, for example.

For this analysis, diverse datasets were employed, incorporating various concept drift detection algorithms. To ensure a more comprehensive analysis, the experimentation was extended to two distinct classifiers that do not incorporate techniques for addressing or adapting to concept drift occurrences.

Tables 3, 4, and 5 exhibit the behavior of the machine learning model when applying different concept drift treatment techniques upon their identification. In Tables 3 to 5, the analysis employs the HT classifier, and the utilized datasets are SEA-abrupt, SEA-gradual, and the Electricity Market, respectively.

The subsequent Tables 6, 7, and 8 adhere to the same analytical approach, altering the datasets to SEA-abrupt, SEA-gradual, and the Electricity Market, respectively. The only distinction lies in the classifier employed, where the KNN (K-Nearest Neighbors) classifier is used.

This classifier variation aims to assess whether different outcomes in concept drift treatment methods could be observed based on the classifier model used. It is important to note that in this analytical phase, the classifiers do not incorporate concept drift treatment techniques. This deliberate choice ensures the reliability of the evaluated methods' outcomes.

Naturally, a model that does not implement concept drift treatment techniques, effectively ignoring them, should have its comparative analysis in the tables based on the metrics of the (Ignore) treatment method. This approach allows for an assessment of the impact of other techniques. As indicated in all tables, batch training emerges as a robust training method in scenarios affected by concept drift.

The reference metrics employed in the analysis are derived from the method that disregards concept drift, thereby establishing a benchmark against which other treatment methods are compared. This provides a discernible framework to assess the efficacy of the alternative approaches.

Notably, in none of the instances did the exclusion of samples significantly influence the model's performance. This consistency indicates that removing samples does not necessarily yield discernible benefits or detriments to the model's stability.

TABLE 7. Accuracy of the models when performing different treatment methods on the SEA-gradual dataset (KNN).

Handling methods	ADWIN	DDM	EDDM	HDDM_A	PageHinkley
Ignore	0.869422	0.869422	0.869422	0.869422	0.869422
Delete	0.869422	0.869422	0.869422	0.869422	0.869422
Retrain	0.869422	0.869421	0.869422	0.869422	0.869422
Batch training	0.886149	0.886149	0.886149	0.886149	0.886149

TABLE 8. Accuracy of the models when performing different treatment methods on the Electricity Market dataset (KNN).

Handling methods	ADWIN	DDM	EDDM	HDDM_A	PageHinkley
Ignore	0.765779	0.765779	0.765779	0.765779	0.765779
Delete	0.765779	0.765779	0.765779	0.765779	0.765779
Retrain	0.76571	0.765757	0.765801	0.765757	0.7658236
Batch training	0.88140	0.88140	0.88140	0.88140	0.88140

Regarding the model retraining strategy, it is observed that the outcomes exhibit a degree of variability. This is an anticipated outcome given the dynamic nature of concept drift and its varying impact on the model based on the evolving data patterns.

The recurring instances of accuracy improvement resulting from batch training are noteworthy. This underscores the potency of processing larger data volumes in segments, allowing the model to adapt more adeptly to changing data distributions.

The exploration of combining these methods within ensembles is particularly intriguing. While the impact of individual techniques may not be markedly pronounced, the amalgamation of diverse strategies could potentially overcome their individual limitations, leading to a more robust and consistently performing approach. This aligns with the recognition that the synergy of methods in ensembles might offer a valuable avenue for effective concept drift management.

The comprehensive analysis of concept drift treatment methods elucidates how distinct strategies influence model accuracy. Moreover, the exploration of ensembles hints at a promising avenue for future investigations. This enriched understanding of effectively addressing concept drift within continuous data streams is a pivotal contribution to advancing the domain of stream learning methodologies.

Given the imperative to comprehensively evaluate more comprehensive treatment models that intrinsically offer a synthesis of the individually analyzed methods, the second segment of this analysis aims to assess the performance of machine learning models–one devoid of concept drift adaptation techniques, and the other endowed with such mechanisms.

This investigation spans across eight distinct datasets, ensuring a comprehensive examination of an array of concept drift occurrences. The classifiers under scrutiny encompass HT, HAT, and EFDT, with HAT being the sole classifier equipped with concept drift handling techniques. The results of this assessment are encapsulated in Figure 1.

This endeavor arises from the recognition that a comprehensive evaluation of model performance necessitates considering a spectrum of potential scenarios encompassing various concept drift manifestations. The contrasting approaches of models that accommodate concept drift and those that do not provide valuable insights into the utility of such adaptation mechanisms.

Enhancing the previous analysis of methods for handling concept drift and broadening the scope to encompass classifiers integrating concept drift adaptation techniques, the outcomes presented in Figure 1 underscore the superior performance of classifiers employing concept drift handling methods.

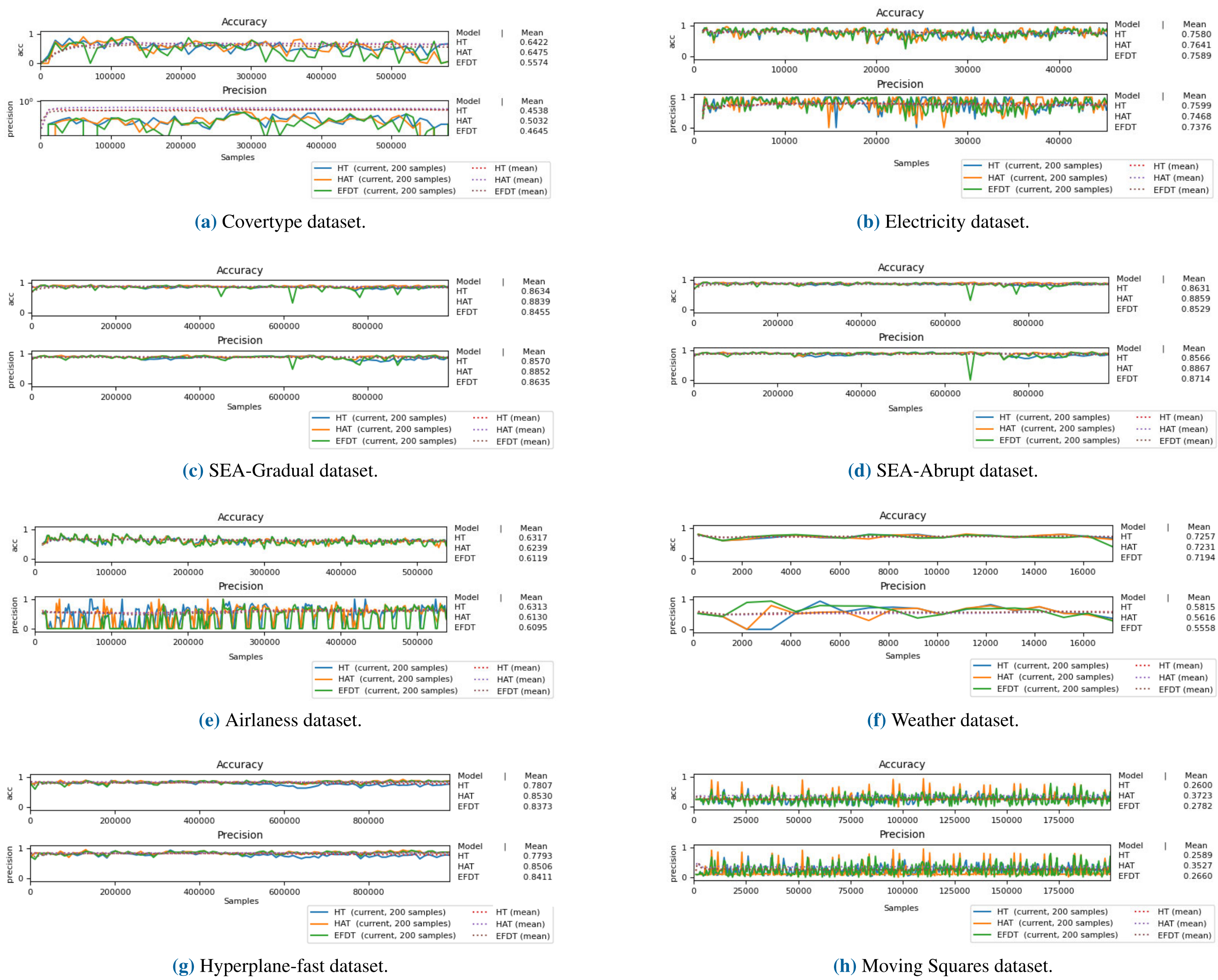
Given the susceptibility of the datasets to concept drift, the algorithm yielded results as anticipated, achieving elevated metrics including accuracy and precision, in contrast to models disregarding the effects of drift.

Notably, in just one of the assessed datasets, the HAT algorithm exhibited slightly inferior metrics. This highlights the notion that the selection of a classification algorithm can lead to varied behaviors.

Consequently, the subsequent analytical phase will encompass the evaluation of diverse classifier types, to unveil distinct behaviors within these classifier categories. Our next experiment shown in Figure 2 illustrates the comparison of the effectiveness of various classification algorithms implementing concept drift treatment techniques across different datasets.

Analyzing the metrics obtained for the batch training method presented from the table 3 to 8, it is evident that, in all results, it was the superior method in terms of accuracy. It is important to note that, depending on the type of concept drift, this method may not be the most suitable.

Considering batch training and analyzing the results presented in Figure 2, the BIC classification algorithm stands out as an example of a method based on batch



(a) Covertype dataset. (b) Electricity dataset. (c) SEA-Gradual dataset. (d) SEA-Abrupt dataset. (e) Airlaness dataset. (f) Weather dataset. (g) Hyperplane-fast dataset. (h) Moving Squares dataset.

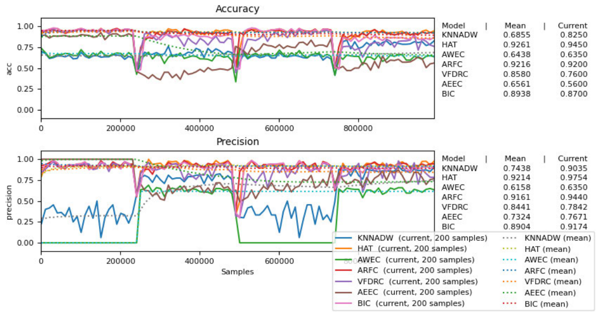
FIGURE 1. Comparison of classification models in data streams with and without concept drift handling across multiple datasets.

training. According to the data presented in the tables, it is evident that this algorithm consistently achieved one of the best performances across all test datasets, regardless of the type of concept drift affecting the data. Although it was not the top-performing algorithm among those analyzed, it consistently demonstrated excellent metrics, highlighting its robustness and reliability. According to [10], ensemble training is one of the most used approaches. As a result of this work, it is observed that the combination of this ensemble method with batch training (BIC) algorithm achieved good performance across all analyzed datasets under different types of concept drift.

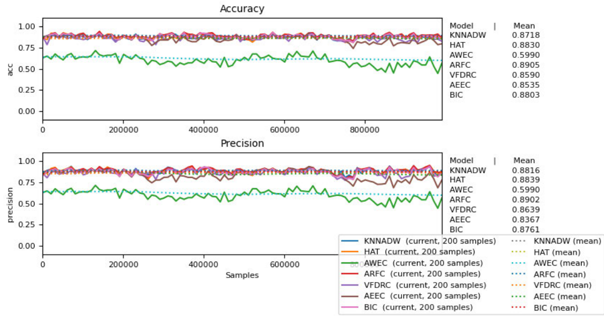
Although ensemble learning is widely used [10], it cannot be guaranteed that its superiority is specifically due to being an ensemble. In the analysis presented here, the superiority is indeed attributed to the ensemble; however, when comparing it to the methods analyzed individually, as shown in the tables, it can be concluded that batch training plays a crucial role in this performance. To further substantiate this observation, we can analyze the AEEC algorithm, which is based on an ensemble but shows highly variable performance. Specifically, it did not perform well in scenarios involving gradual concept drift, while it demonstrated good performance in abrupt concept drift. This contrasts with the complementary nature of batch training combined with ensemble training (BIC), which performs favorably regardless of the type of concept drift analyzed.

Other algorithms, leveraging classification techniques and treatments more tailored to the types of concept drift and data distribution, have demonstrated superiority. However, irrespective of this, algorithms based on batch training consistently exhibited commendable metrics.

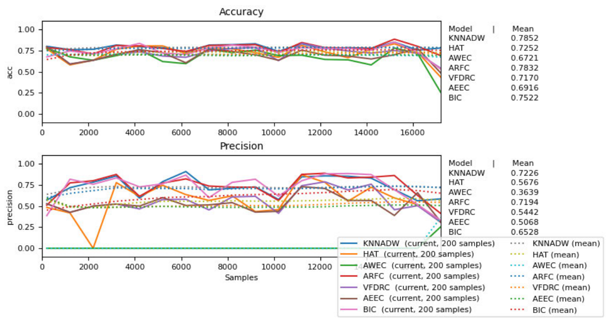
A more in-depth analysis of the results about the SEA-A dataset in Figure 2a, characterized by abrupt concept drifts, reveals that decision tree-based algorithms exhibited superior



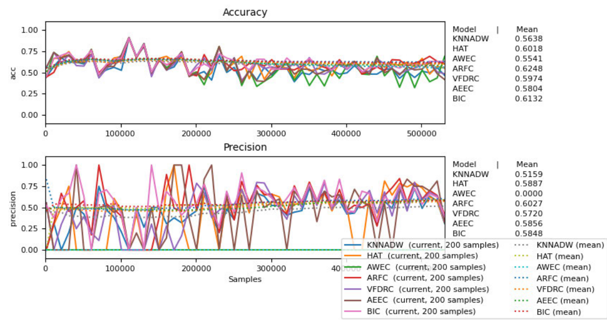
(a) SEA-Abrupt dataset.



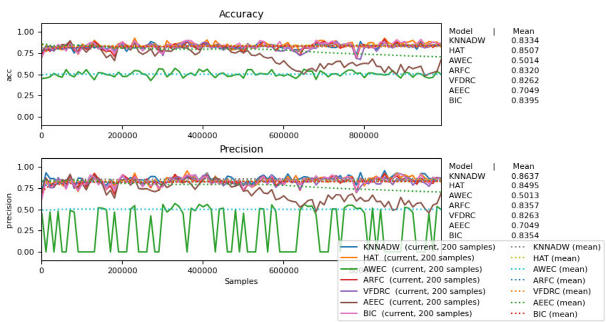
(b) SEA-Gradual dataset.



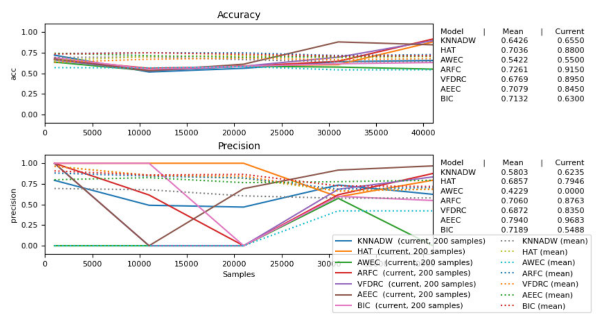
(c) Weather dataset.



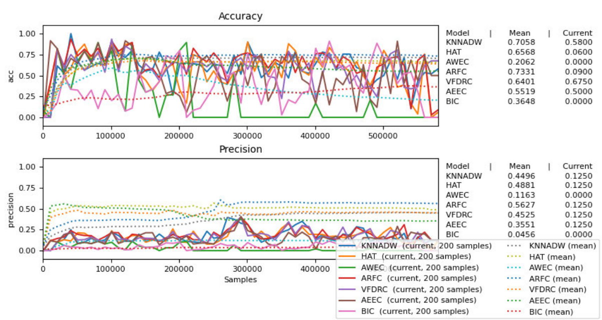
(d) Airlaness dataset.



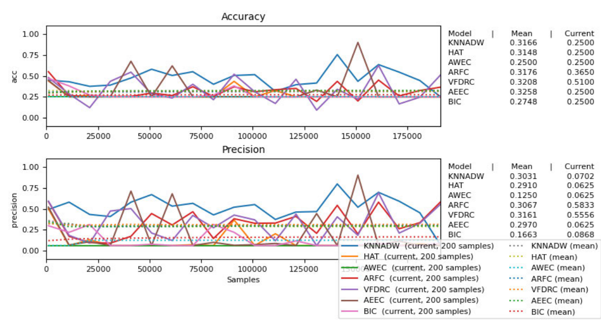
(e) Hyperplane-fast dataset.



(f) Electricity dataset.



(g) Covertype dataset.



(h) Moving Squares dataset.

FIGURE 2. Comparison of the performance among different classification algorithms for data affected by concept drift.

performance. As detailed in the definition of HAT algorithm, the operation involves applying concept drift detection at each branch of the decision tree. Naturally, we can infer that the model can adapt more swiftly, achieving higher efficacy, because, with this treatment approach, in the case of abrupt concept drifts, the affected branch is promptly addressed without impacting other branches, thus avoiding training delays.

This analysis is consistent when examining the SEA-Gradual dataset shown in Figure 2b, characterized by gradual concept drifts and a longer window for data observation. Algorithms such as KNNADW, with a more extended window, allow for more accurate identification of this type of deviation when compared to decision tree-based algorithms. This is due to the analysis in each branch being more time-consuming before the deviation is accurately identified.

Another noteworthy algorithm is VFDRC, which is based on decision trees and does not implement methods to circumvent concept drifts once identified. However, it is an incremental training algorithm that naturally adapts the model to new incoming data. Once again, depending on the type of concept drift occurrence, it may not deliver the best performance. Nevertheless, upon analyzing the results obtained in Figure 2, it demonstrated remarkable metrics in all the analyzed datasets.

IV. CONCLUSION

This study conducted a comprehensive evaluation of classification algorithms implementing concept drift treatment, analyzing their performance across various datasets and scenarios. By systematically applying different concept drift detection algorithms and exploring diverse forms of drift, this work provides an empirical benchmark that highlights critical insights for practitioners and researchers.

One significant finding is the effectiveness of a training method that employs a sliding data window, leading to notable performance improvements without altering the data. This approach was consistent across various classification and detection methods, offering a versatile solution for dynamic data environments. Additionally, while exclusion techniques did not enhance accuracy, they demonstrated the potential for reducing storage and computational costs without compromising performance–an essential consideration for resource-constrained systems.

The study also revealed that combining these strategies within ensemble frameworks can amplify their strengths, yielding higher adaptability and robustness in dynamic settings. These results underscore the practical implications of choosing appropriate concept drift handling techniques based on specific application needs.

This comparative analysis not only sheds light on the nuances of existing methods but also bridges the gap between theoretical advancements and practical applications. By providing a detailed exploration of algorithm performance and trade-offs, the study equips practitioners with actionable insights to guide their selection of methods tailored to the nature of their datasets and drift scenarios.

Ultimately, the findings reaffirm the critical importance of addressing concept drift in machine learning. Models that incorporate effective drift management techniques consistently outperform static counterparts, achieving greater accuracy and adaptability over time. This work lays a foundation for future research aimed at optimizing and extending these methods, emphasizing their pivotal role in creating resilient machine learning systems for ever-evolving real-world data environments.

V. FUTURE WORKS

Our study conducted an in-depth analysis of the behavior of various approaches to handling concept drift using different classifiers. One potential direction for future work is to extend this analysis to regression problems, which would expand the scope of our findings and their applicability to a broader range of scenarios. Furthermore, evaluating additional classification algorithms not covered in this study could provide further insights and contribute to a more comprehensive understanding of concept drift handling techniques.

Building on the analyses and observations made in this study, another promising avenue for future research lies in the development of specialized algorithms for handling concept drift. These algorithms could be designed to generalize their effectiveness regardless of the type of concept drift encountered, thereby addressing the limitations observed and contributing to advancements in the field. Additionally, future work could explore how these methods perform when extended to handle concept drift in deep learning models and how they adapt to dynamic environments, further enhancing their applicability in real-world scenarios.

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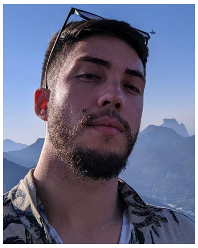

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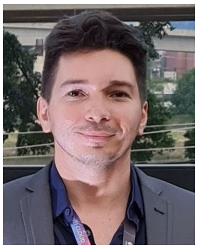

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